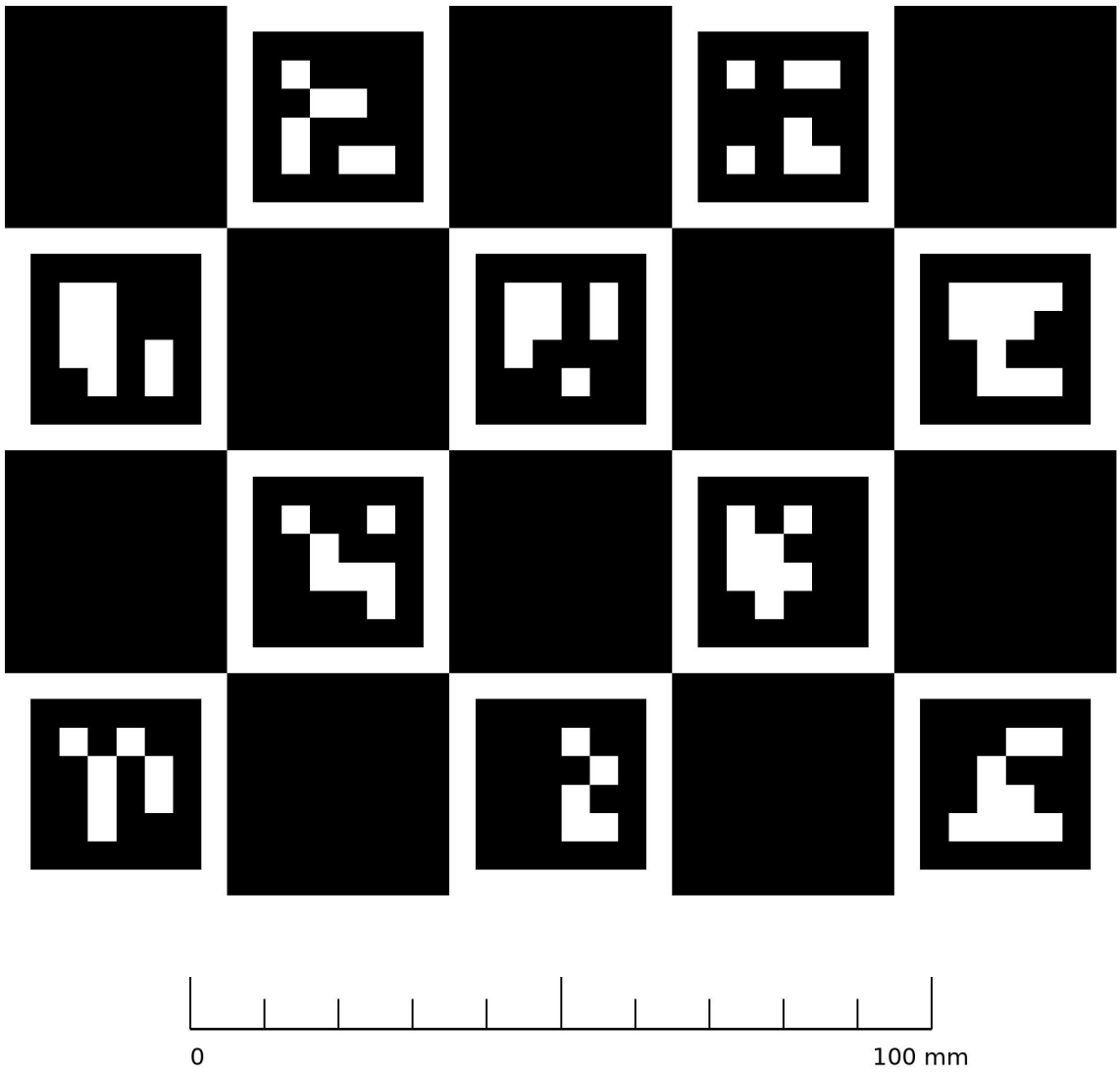
SO101_PLATE table board: ChArUco 5 x 4

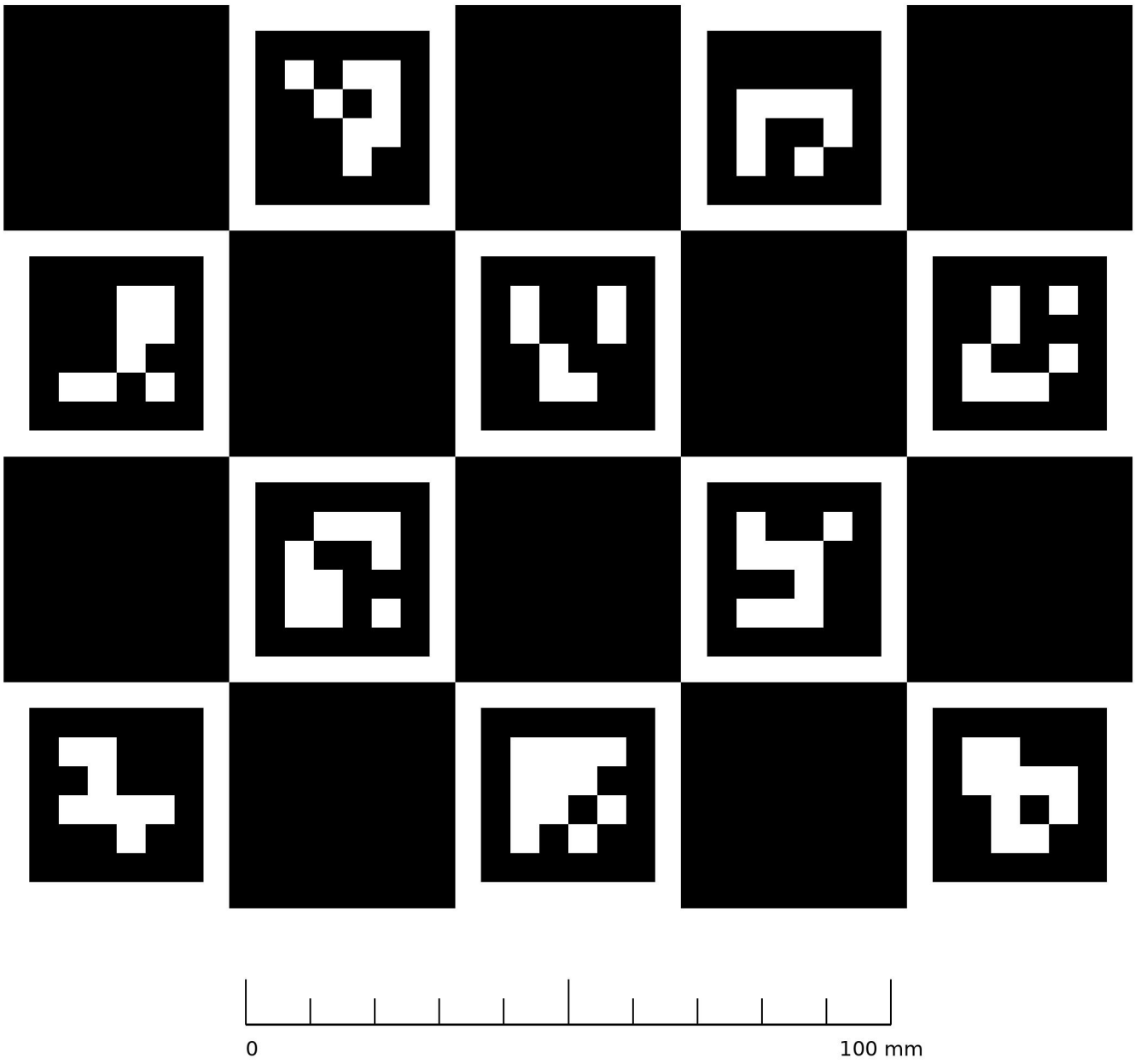
For the WRIST D405's hand-eye calibration (lies flat on the desk).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.
Dictionary DICT_4X4_50, first marker id 20.



CHECK 1: the scale bar measures exactly 100 mm (else the printer scaled: reprint).
CHECK 2: measure across 4 squares with calipers / a steel ruler:
  expected 4 x 30.0 = 120.0 mm (markers 23.0 mm).
Enter the measured square size (m) as table_board.square_m and marker_m (scaled the same)
in config/calibration_targets_so101_plate.yaml.

SO101_PLATE hand board: ChArUco 5 x 4

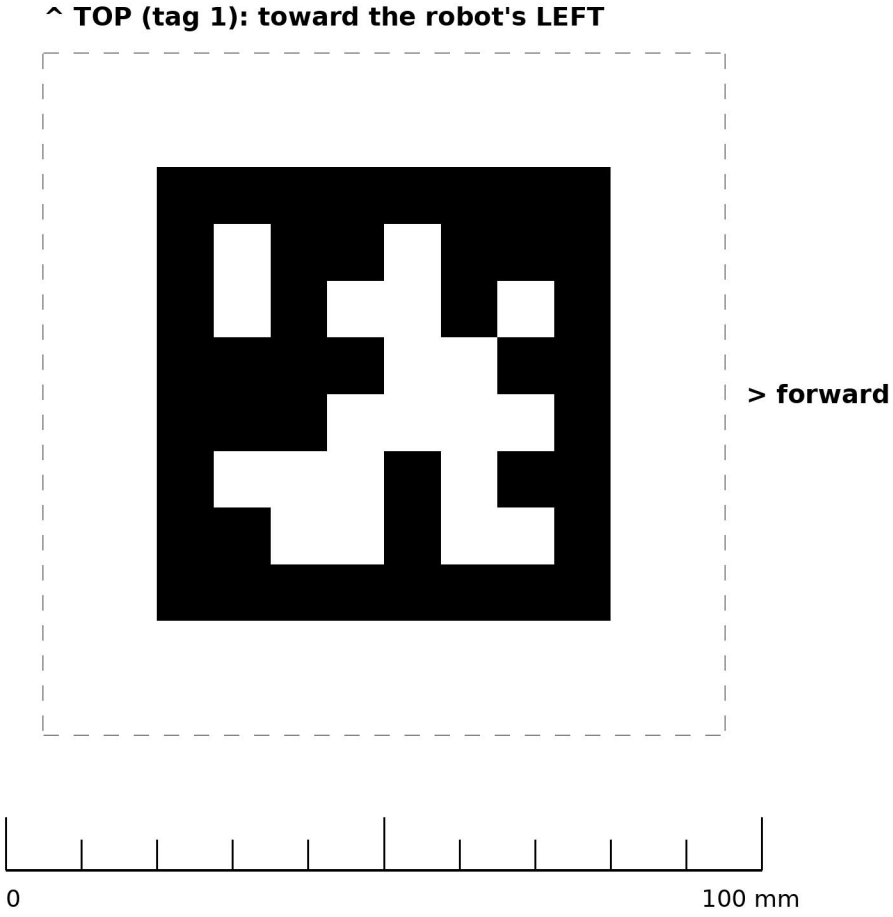
For the fixed D435i's calibration (held in the SO-101's jaw), later.
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.
Dictionary DICT_4X4_50, first marker id 0.



CHECK 1: the scale bar measures exactly 100 mm (else the printer scaled: reprint).
CHECK 2: measure across 4 squares with calipers / a steel ruler:
  expected 4 x 35.0 = 140.0 mm (markers 27.0 mm).
Enter the measured square size (m) as hand_board.square_m and marker_m (scaled the same)
in config/calibration_targets_so101_plate.yaml.

SO101_PLATE AprilTag 1 (DICT_APRILTAG_36h11)

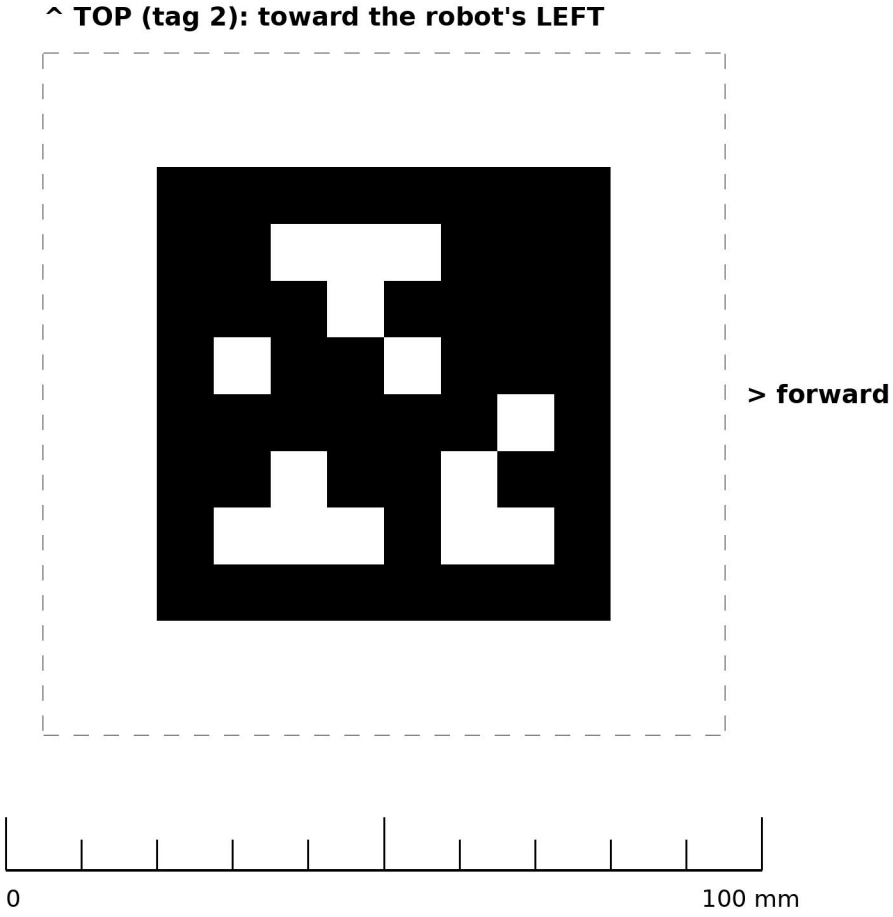
The drift monitor's fixed reference on the desk (outside the work band).
Place at view far-left (behind base): centre x -3.0 cm / -1.18 in,
y 22.0 cm / 8.66 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 2 (DICT_APRILTAG_36h11)

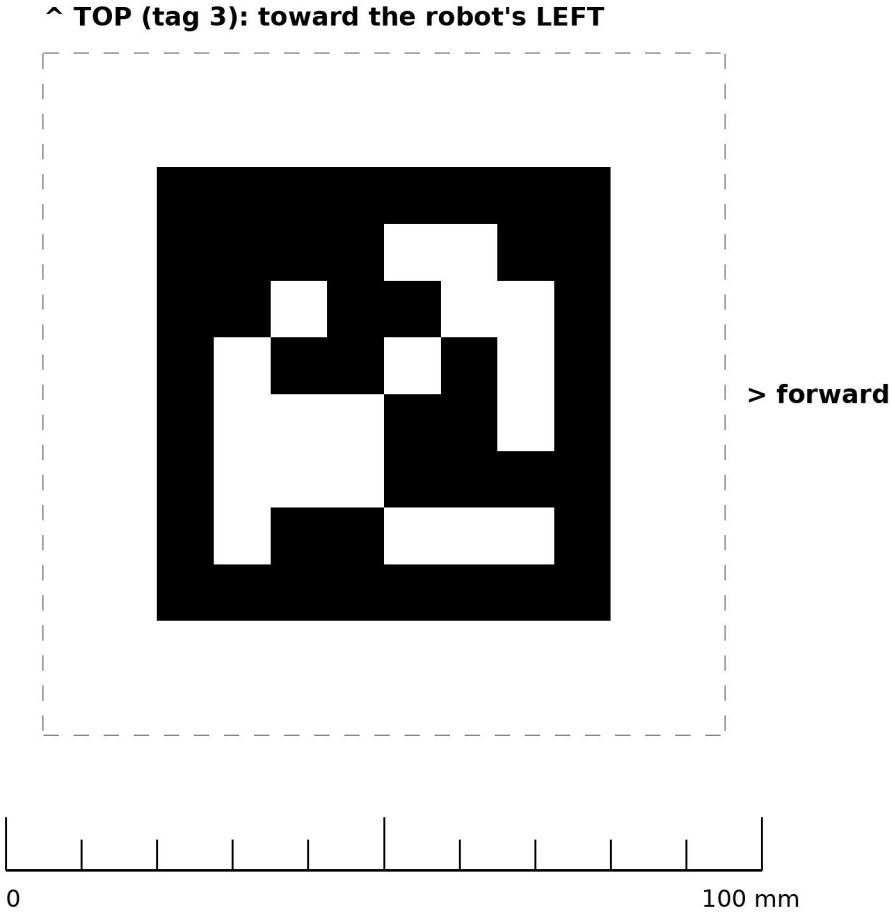
The drift monitor's fixed reference on the desk (outside the work band).
Place at view far-right (behind base): centre x -4.5 cm / -1.77 in,
y -27.0 cm / -10.63 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 3 (DICT_APRILTAG_36h11)

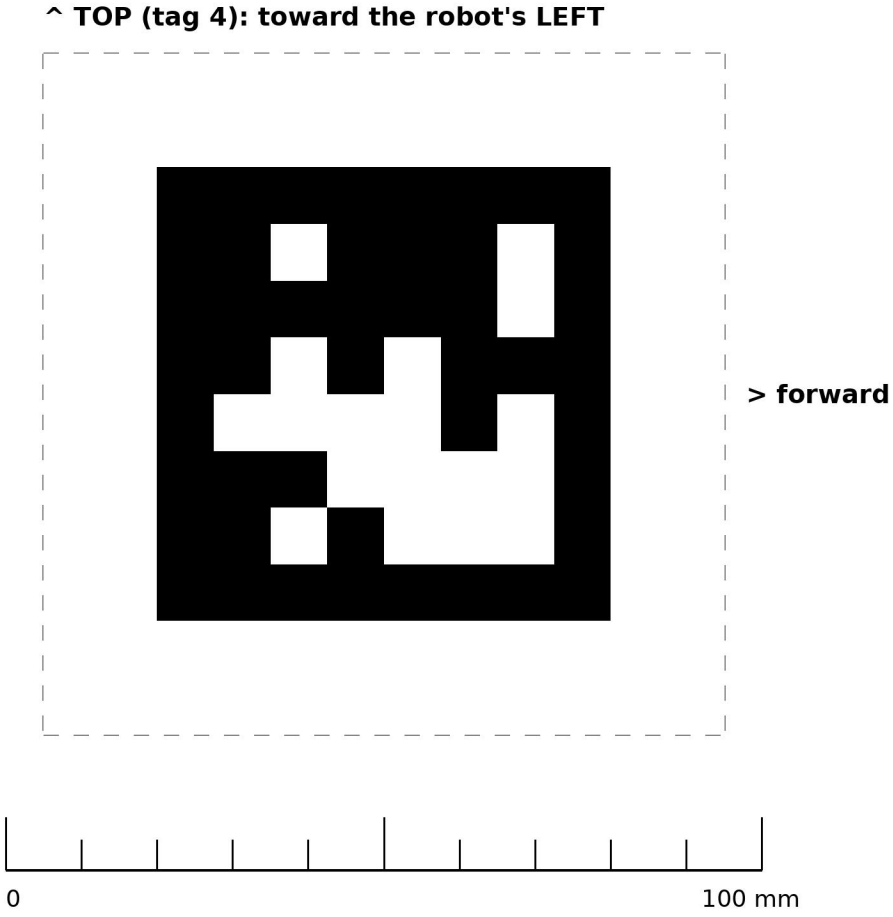
The drift monitor's fixed reference on the desk (outside the work band).
Place at beside base left: centre x 0.0 cm / 0.00 in,
y 12.0 cm / 4.72 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 4 (DICT_APRILTAG_36h11)

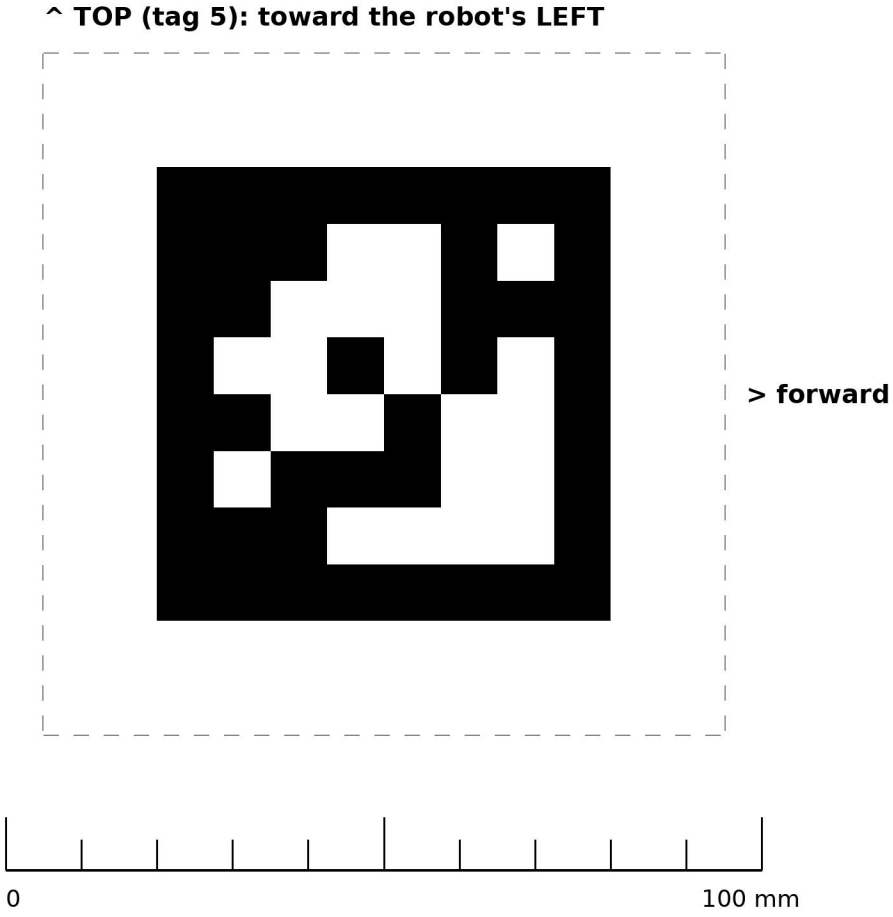
The drift monitor's fixed reference on the desk (outside the work band).
Place at beside base right: centre x 0.0 cm / 0.00 in,
y -13.0 cm / -5.12 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 5 (DICT_APRILTAG_36h11)

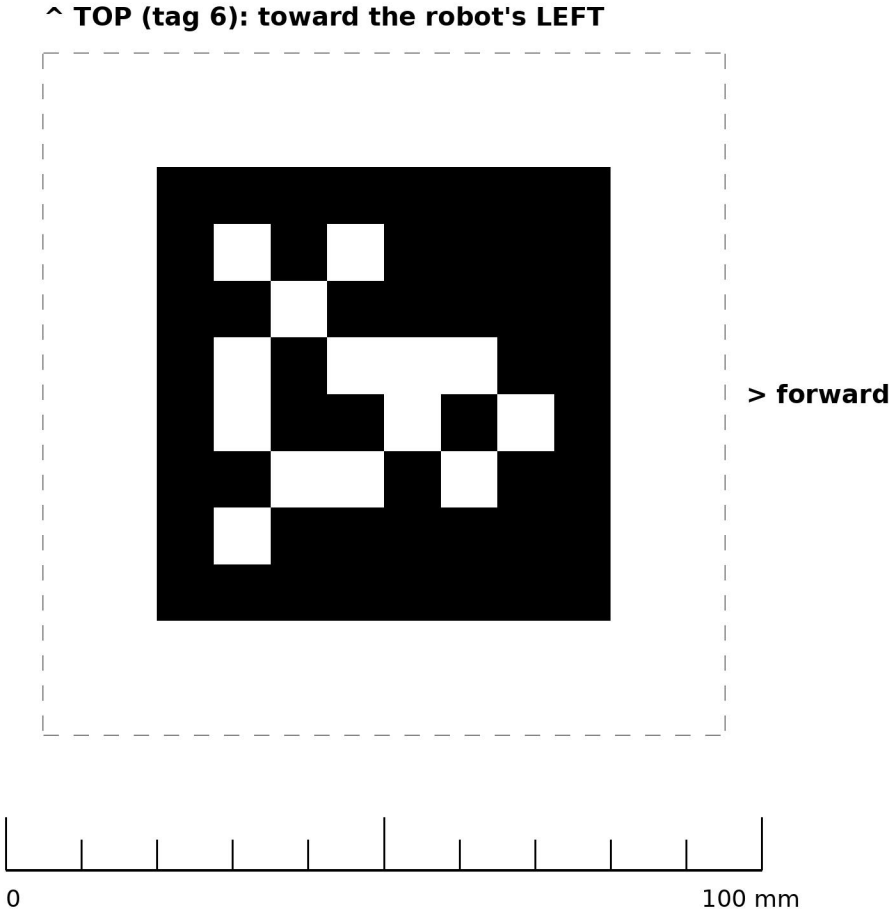
The drift monitor's fixed reference on the desk (outside the work band).
Place at view left side: centre x 11.0 cm / 4.33 in,
y 18.5 cm / 7.28 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 6 (DICT_APRILTAG_36h11)

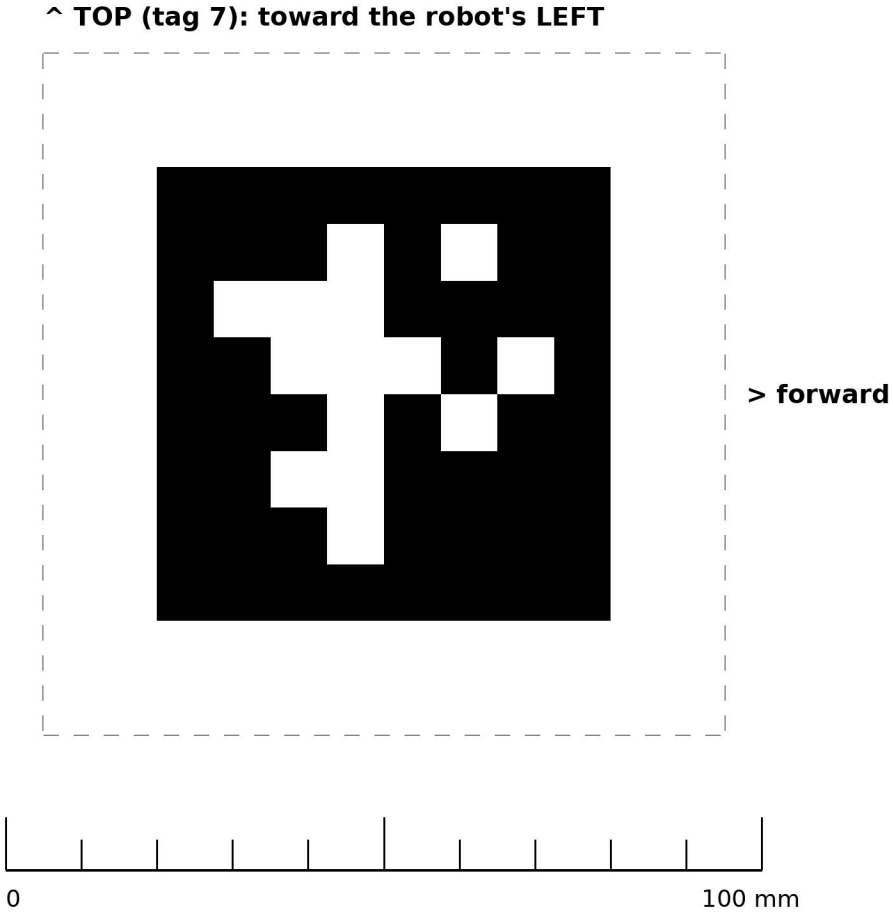
The drift monitor's fixed reference on the desk (outside the work band).
Place at view right side: centre x 11.0 cm / 4.33 in,
y -21.5 cm / -8.46 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 7 (DICT_APRILTAG_36h11)

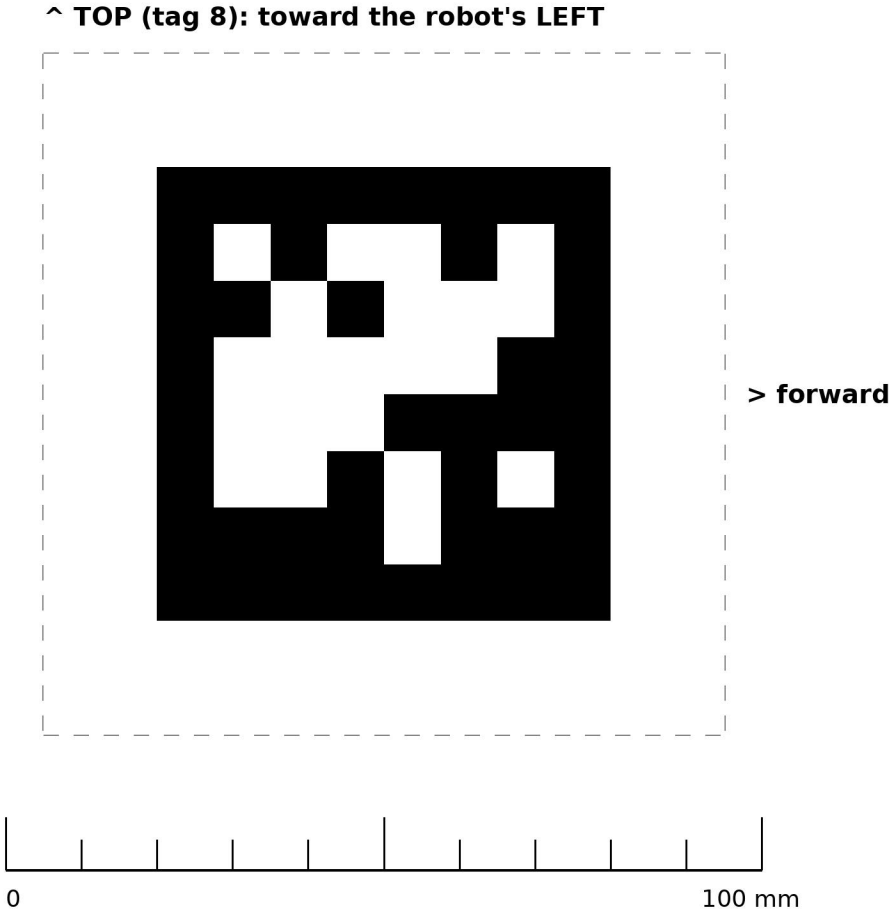
The drift monitor's fixed reference on the desk (outside the work band).
Place at view near-left: centre x 30.0 cm / 11.81 in,
y 12.5 cm / 4.92 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 8 (DICT_APRILTAG_36h11)

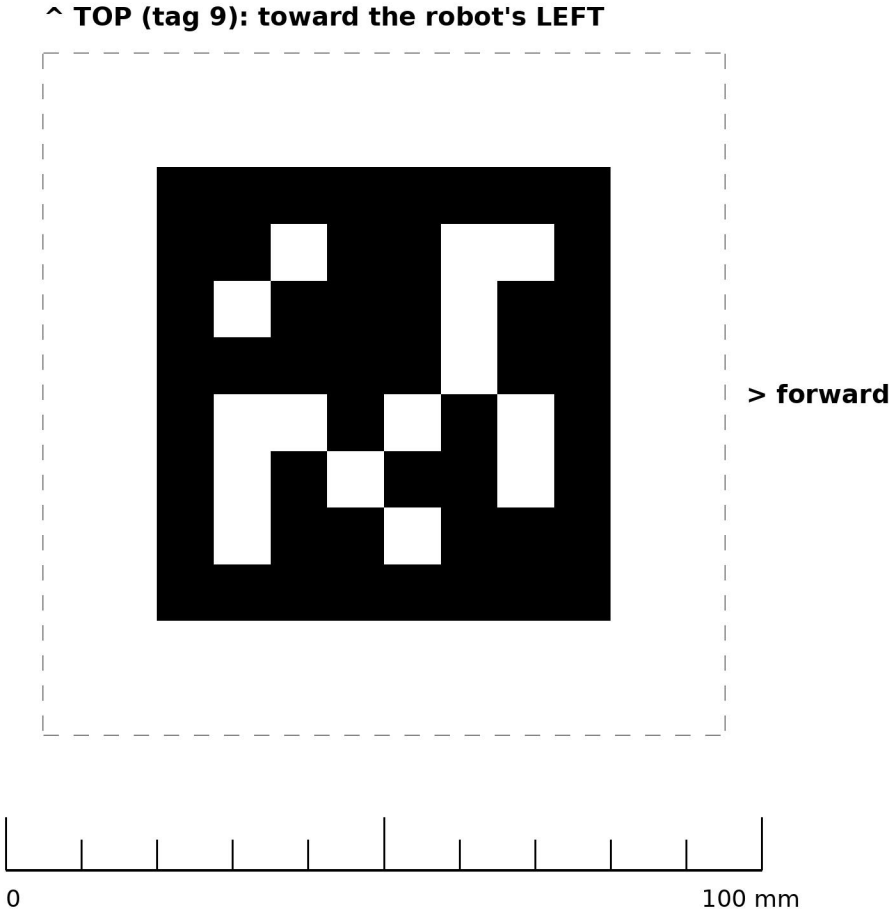
The drift monitor's fixed reference on the desk (outside the work band).
Place at view near-right: centre x 30.0 cm / 11.81 in,
y -15.0 cm / -5.91 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.

SO101_PLATE AprilTag 9 (DICT_APRILTAG_36h11)

The drift monitor's fixed reference on the desk (outside the work band).
Place at view near-centre: centre x 33.0 cm / 12.99 in,
y 0.0 cm / 0.00 in from the robot base (x forward, y left).
Print at 100 % / 'Actual size' (NOT 'Fit to page'), on matte paper, best printer quality.
Glue it FLAT to something rigid (foam board, 3 mm plywood, acrylic). No wrinkles, no lamination glare.



CHECK: the scale bar measures 100 mm; the black square measures 60.0 mm.
Enter the measured size as apriltags.size_m in config/calibration_targets_so101_plate.yaml.
Cut on the dashed line (keep the white border: the detector needs it). Write the id on the back.
On the board: the TOP edge points to the robot's LEFT (+y), the RIGHT edge points away from
the robot (forward, +x). Position within ~1-2 cm is fine (it gets measured); flat + this turn matter.